

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Master of Computer Applications (MCA)

Semester-I University Examination March-2018

CA717 Fundamentals of Data Communications

Date: 26.03.18, Monday

Time: 01:30 p.m to 04:30 p.m

Maximum Marks: 70

Instructions:

- 1. Attempt both sections in separate answer books.**
- 2. Figure to right indicates marks.**
- 3. Make suitable assumption when necessary.**

SECTION I

Q1. (A) Answer Following.

1. Define following terms: 1) Encoding 2) Modulating. **[02]**
2. True/False: UDP is connection oriented protocol. **[01]**
- 3.. Give full form of: 1) ISO 2) IEEE. **[02]**

Q1. (B) Answer Following.

1. Give full form of OSI. What is the functioning of presentation layer? **[06]**
2. What is error control? Explain with an example.
3. List down any 4 applications of computer network.

Q2. Answer Following.

1. What are different networking topologies? Explain each one in brief. **[12]**
2. Write a short note on LAN and explain its working in detail.

OR

Q2. Answer Following.

1. Write a short note on WAN and explain its working in detail. **[12]**
2. Write a short note on transmission modes.

Q3. Answer Following.

1. Explain circuit switching, packet switching and message switching techniques. **[12]**
2. Write short note on MAN and explain its working in detail.

OR

Q3. Answer Following.

1. Give overview of OSI model. **[12]**
2. Write a short note on CRC.

SECTION II

- Q4. (a) Do as Directed**
1. Define following: 1) Cryptography 2) Plain text 3) Cipher text. [03]
 2. What is subnetting? [01]
 3. What is IEEE project 802? [01]

- Q4. (b) Answer Following.** [06]
1. What is ATM? What are its applications?
 2. Define following: 1) Encryption 2) Decryption.
 3. What is IP? What are its use?

- Q5. Do as directed** [12]
1. Write a short note on switch.
 2. Write short note on bridge.

OR

- Q5. Answer Following.** [12]
1. Explain working of gateway.
 2. Explain working of router.

- Q6. Answer Following** [12]
1. Explain substitution cipher with an example.
 2. What is UDP? Explain how UDP works.

OR

- Q6. Answer Following.** [12]
1. What is FTP? Explain how FTP works.
 2. What is TCP? Explain how TCP works.